

Solution Architecture

[illegible]

Maximum Marks: 5 Marks

Architecture Flow

Streamlit Presentation Layer ↔ FastAPI API Layer ↔ Gemini AI Service; Streamlit also connects to local formatter modules for TXT/DOCX/PDF export. Configuration and secrets are supplied through environment variables.

Component Name	Description / Role	Technology Used
Presentation Layer	Collects inputs, displays preview and provides downloads.	Streamlit
API Layer	Exposes /generate and /health endpoints.	FastAPI + Uvicorn
AI Service	Generates document text from user inputs.	Google Gemini API
Validation	Validates document request fields.	Pydantic
Formatting	Creates DOCX/PDF and HTML preview.	python-docx, fpdf2, HTML
Configuration	Loads API key, model and server settings.	.env / python-dotenv